

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Simplify using K-map. & realize using Logic Gates.  
(CO2)

$$F(A,B,C,D)=\Sigma m(0,1,2,3,5,7,13,15)$$

**Q.24** Explain SIPO Shift register in detail with the help of neat diagram. (CO3)

Q.25 Explain the Architecture of 8051 microcontroller.  
(CO5)

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### 3rd Sem / Artificial Intelligence & Machine Learning

**Subject : Digital Electronics & Microcontrollers**

Time : 3 Hrs.

M.M. : 60

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The number of digits in chemical number system is (CO1)

- a) 8                      b) 10  
c) 16                     d) 2

Q.2 The NOR gate is AND gate followed by (CO1)

- a) Not Gate                      b) OR Gate
- c) NOR Gate                    d) EXOR Gate

Q.3 In 16:1 MUX, How many select lines are required.  
(CO2)

- a) 4                                      b) 3  
c) 2                                      d) 1

Q.4 A Half adder circuit has \_\_\_\_\_ inputs. (CO3)

- a) 2                                      b) 3  
c) 4                                      d) 8

Q.5 The basic storage element in digital systems is \_\_\_\_\_ (CO3)

- a) Counter                              b) Encoder  
c) Flip Flop                              d) Mux

Q.6 When the microcontroller executes some arithmetic operations, then the flag bits of which register are affected? (CO5)

- a) PSW                                      b) SP  
c) DPTR                                      d) PC

### SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Write any two differences between digital signal and analog signal. (CO1)

Q.8 Draw the symbol of OR Gate. (CO1)

Q.9 2'S Complement of 10110 is \_\_\_\_\_ (CO1)

Q.10 Full form of SIPO \_\_\_\_\_ (CO3)

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Q.11 What is the size of ROM in 8051 microcontroller. (CO5)

Q.12 A.A= \_\_\_\_\_ (CO2)

### SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Convert  $(47)_8 = (?)_2 = (?)_{16}$  (CO1)

Q.14 Solve =  $\begin{array}{r} 101011 \\ -10001 \\ \hline \end{array}$  (Co1)

Q.15 Write logic expression for AND gate and draw its symbol and truth table. (CO1)

Q.16 Solving using Boolean algebra (CO1)

$$Y = A\bar{B}C + ABC + AC + A\bar{C}$$

Q.17 Write a short note on 1:4 DeMUX. (CO2)

Q.18 Explain Full Adder with diagram. (CO3)

Q.19 Differentiate between Latch and Flip Flop. (CO2)

Q.20 Explain D Flip Flop. (CO3)

Q.21 Draw Pin diagram of 8051 microcontroller. (CO5)

Q.22 Write a short note on RAM structure of 8051 microcontroller. (CO5)

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